Today's Agenda

- Game theory real world paradoxes : the **Braess paradox**.
- Back to simultaneous-move games of incomplete information : the **free riding game**.
- In preparation to next lecture : **conditional (Bayesian) probability** with examples.

The Braess Paradox

Definition

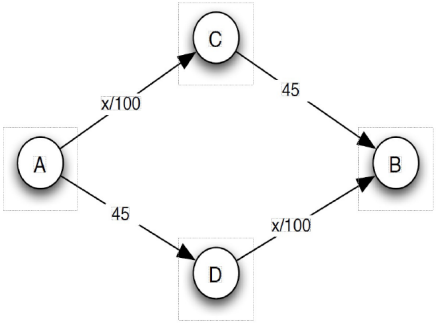
BRAESS PARADOX
N-player, simultaneous-move game (complete information) modeling a transportation network.
Paradox : **adding a new, very fast route to a transportation network can increase traffic and transportation time for everybody**.

Example

Real-world cases

- **Stuttgart, 1969** : shortly after D. Braess published his theory, the city invested heavily in its downtown road network by building a highly efficient new road segment. Traffic immediately ground to a halt. When the city closed the newly built road section, traffic immediately returned to normal.
- **NYC, 1990** : 42nd street (one of Manhattan's busiest crosstown arteries) closed on "Earth day". Media outlets warned of a "traffic apocalypse", but when the street closed, the surrounding gridlock dropped drastically, and traffic moved faster.
- **Seoul, South Korea, 2003** : the destruction of a six-lane highway to build a park actually improved travel time into and out of the city. Traffic volume stayed roughly the same before and after.

Braess – Initial Situation



- **N = 4000 cars** (players). Two **directed** routes to go from A to B.
- Travel time on an "edge" of the graph (e.g. edge AC) is a function of the number of cars : $T_e = T_e(x)$.
- Payout for each car is $-T_e(x)$: each car's objective is to **minimize** travel time.

Is there a NE in which all cars take the same route ? No. If all cars went through C (route AC+CB), travel time would be $4000/100 + 45 = 85$ minutes. A car would have an incentive to switch to the route through D (AD+DB) : $1/100 + 45 = 45.01$.

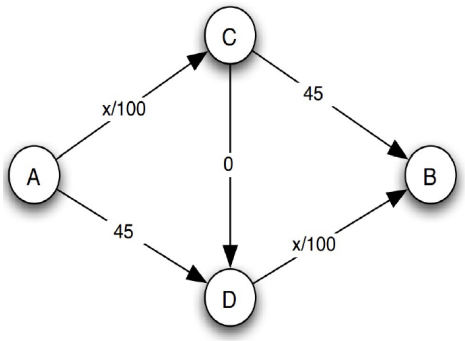
Key idea

For different cars to take different routes (still a **pure-strategy NE**), since all cars are identical, with the same payout, travel time should be the same on the two routes *in the limit $N \to \infty$, in which a single car cannot affect the travel time of a route.*
2000 cars on each route is a NE, with travel time $2000/100 + 45 = 65$ minutes for each car.

Example

Odd number of cars : e.g. 7 cars, 3 on AC+CB and 4 on AD+DB. Is this a NE ? Yes, because even for the cars on AD+DB, switching to AC+CB would leave travel time unchanged.

Braess – New Route is Added



New, superfast edge CD (can only go from C to D). Three possible routes from A to B now : AC+CB ; AD+DB ; AC+CD+DB.

- As before, everybody taking AC+CB, or everybody taking AD+DB, cannot be a NE.
- **Everybody on AC+CD+DB** : travel time $= 4000/100 + 0 + 4000/100 = 80$. If somebody switched to AC+CB, travel time would be $4000/100 + 45 = 85$; the same if somebody switched to AD+DB. So **AC+CD+DB is a NE!**
- **Half on AC+CB, half on AD+DB** (65 minutes) is not a NE anymore : if one car switches to AC+CD+DB, travel time would be $2001/100 + 0 + 1999/100 \simeq 40$ minutes.

Key idea

No matter the strategy of the other players, switching from AC+CD+DB to another route involves switching from an edge with travel time $x/100$ (at most 40) to an edge with travel time 45.
AC+CD+DB is the only NE! Travel time is 80, **15 minutes longer** than the NE before the route was added.

A Pure-Strategy NE for Every Network

Definition

POTENTIAL ENERGY
Potential energy of an edge e when x_e cars travel on this edge :

$$E_e(x_e) = T_e(1) + T_e(2) + \cdots + T_e(x_e)$$

Potential energy of a network $\mathcal{N}$: the sum of the potential energy of all its edges :

$$E_{\mathcal{N}} = \sum_{e \in \mathcal{N}} E_e(x_e)$$

Warning

Potential energy is different from the aggregate travel time (total time spent on this edge by the cars traveling on it), which is $x\,T_e(x)$.

Key idea

If one single car leaves the network, potential energy decreases by exactly the traveling time of this car.

Example

3 cars, one on AC+CB and 2 on AD+DB :

$$E_{\mathcal{N}} = T_{AC}(1) + T_{CB}(1) + (T_{AD}(1) + T_{AD}(2)) + (T_{DB}(1) + T_{DB}(2))$$

One car traveling on AD+DB leaves the network :

$$E_{\mathcal{N}} = T_{AC}(1) + T_{CB}(1) + T_{AD}(1) + T_{DB}(1)$$

The difference is $T_{AD}(2) + T_{DB}(2)$: the traveling time of this car.
If this car re-enters on another route, e.g. AC+CB, the potential energy increases by its traveling time on the new route, $T_{AC}(2) + T_{CB}(2)$:

$$E_{\mathcal{N}} = (T_{AC}(1) + T_{AC}(2)) + (T_{CB}(1) + T_{CB}(2)) + T_{AD}(1) + T_{DB}(1)$$

Key idea

- When one car switches to a better route (which decreases its traveling time), **potential energy decreases**.
- (Pure-strategy) Nash equilibrium corresponds to a **min of potential energy**.
- Only a finite number of configurations $\rightarrow$ there is necessarily a min of potential energy. So **there is necessarily a pure-strategy NE**.

Finding a NE for an Arbitrary Network

Summary

Pseudocode

1. Start from an arbitrary configuration (strategy profile), compute potential energy.
2. Consider all possible changes of strategy **for 1 player**. If there is any change that decreases potential energy, update the configuration.
3. Stop when no favorable change is found. This is the NE and the minimum of Potential energy.

Recap : Bayesian Games

Definition

BAYESIAN GAME (simultaneous-move game of incomplete information)
Defined by : players, an action set for each player, possible types θ_i for each player i and a probability distribution over player i types $p(\theta_i)$.
A pure strategy is a plan of action **for each type of player**.

Definition

BAYESIAN NASH EQUILIBRIUM (BNE)
$(s_1(\theta_1), \ldots, s_N(\theta_N))$ is a BNE iff, for each player i, her **expected payout** is higher than or equal to her expected payout using any of the other available strategies, keeping fixed the strategies of the other players, which **depend on their types** :

$$\mathcal{P}_i(s_i(\theta_i), s_{-i}(\theta_{-i})|\theta_i) \geq \mathcal{P}_i(\tilde{s}_i(\theta_i), s_{-i}(\theta_{-i})|\theta_i)$$

Take the expected value over all possible types of the other players.

Example

Public good and free riding (see week 04)

$1\backslash 2$	W	N
W	$(1-c_1, 1-c_2)$	$(1-c_1, 1)$
N	$(1, 1-c_2)$	$(0, 0)$

$c_1 = 0.5$ known to everybody; $c_2 = 0.2$ (Efficient type, prob p) or 1.2 (Inefficient type, prob $1-p$), known only to Player 2; p is common knowledge. 'Work' is dominated for the Inefficient type.

- Efficient works, Inefficient does not : Player 1 gets 0.5 if he works, p if he does not $\rightarrow$ works if $p < 0.5$, does not work if $p > 0.5$. For $p > 0.5$: **BNE** (N, WN).
- Player 2 never works : Player 1 works regardless of p : **BNE** (W, NN).

Key idea

- If $p < 0.5$: unique pure-strategy BNE (W, NN).
- If $p > 0.5$: two pure-strategy BNE (W, NN) and (N, WN).

Harsanyi representation (introduced in the last lecture) : one table with the strategy of Player 2 conditional on his type ('WN' = "Work if efficient, Not Work if Inefficient").

1\2	WN	NN	NW	WW
W	(0.5, 1-0.2p)	(0.5, 1)	(0.5, 1.2p-0.2)	(0.5, p-0.2)
N	(p, 0.8p)	(0,0)	(1-p, -0.2(1-p))	(1, p-0.2)

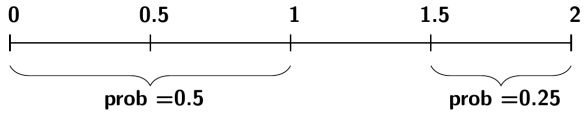
After elimination of the dominated strategy W for the Inefficient Player 2, only WN and NN remain : we find the same pure-strategy equilibria, (W, NN) if $p < 0.5$, (W, NN) and (N, WN) for $p > 0.5$.

Public Good with a Continuum of Types

Same game, but c_1 and c_2 can take a **continuum of values** : for example c_1 and c_2 **uniformly distributed in the interval** $[0, 2]$.

Key formulas

The probability that each of these variables is between x_1 and $x_2 \in [0, 2]$ is the length of the interval $|x_1 - x_2|$ divided by the total length of the domain (here 2).



- Reasonable to expect that when the cost of Player i is low, she will decide to work; when her cost is high, she will not work.
- No player would ever work if her cost is above 1.
- Look for a solution where each player works if her cost is below a **threshold** T_i^* (not necessarily the same for the two players).

Key formulas

Player 2's threshold is T_2^*. What should Player 1 do?

- If she works, her payout is

$$\mathcal{P}_1^W = 1 - c_1$$

- If she does not work, her expected payout is 1 times the probability that Player 2 works, i.e. the probability that $c_2 < T_2^*$:

$$\mathcal{P}_1^N = \frac{T_2^*}{2}$$

Player 1 works if $\mathcal{P}_1^W > \mathcal{P}_1^N$, hence her threshold is

$$T_1^* = 1 - \frac{T_2^*}{2}$$

The same way, Player 2's best response to T_1^* is

$$T_2^* = 1 - \frac{T_1^*}{2}$$

Key idea

The solution is

$$T_1^* = T_2^* = \frac{2}{3}$$

Challenge : Public Good Two Days in a Row

Example

What if we played twice the same game?

- Each day, each player decides whether or not to work. Same payout matrix.
- The costs c_1 and c_2 are known only to the respective player. However, each player's cost is **THE SAME ON BOTH DAYS**.

Would you use the same strategy on both days?

On day 2, you have more information. If my "opponent" worked on day 1, her cost is likely to be low → she is also more likely to work on day 2 → my threshold for working is lower…

Warning

On day one, would you behave the same as in the 1-period game?

Conditional (Bayesian) Probabilities

Definition

CONDITIONAL PROBABILITY
Probability of an event A, conditional on an event B : if B occurs, what's the probability that A also occurs?

Key formulas

Bayes formula :

$$Prob(A|B) = \frac{Prob(A\&B)}{Prob(B)}$$

Key idea

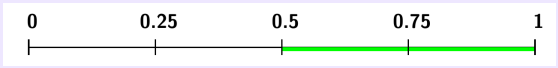
Bayesian inference is a rule to update probabilities when new information becomes available.

Conditional Probability Examples

Example

Example 1
A random variable x is uniformly distributed between 0 and 1 ($x \sim \mathcal{U}([0, 1])$).

- What is the probability that $x > 0.75$? (the *unconditional probability*)
- Now you receive the information that $x > 0.5$. How do you update the probability that $x > 0.75$? (the *conditional probability*)



Solution (not on the slides) :

$$Prob(x > 0.75) = 0.25$$

$$Prob(x > 0.75 \,|\, x > 0.5) = \frac{Prob(x > 0.75)}{Prob(x > 0.5)} = \frac{0.25}{0.5} = 0.5$$

Example

Example 2

— You know that X has two children. What is the probability that she has (at least) a girl ?

— Then you ask her : do you have a boy ? She says : yes.

— What is now the probability that she also has a girl ? a. 1/2, b. 1/3, c. 2/3

Solution (not on the slides) : the four cases BB, BG, GB, GG are equally likely.

$$Prob(\text{girl}) = \frac{3}{4}$$

$$Prob(\text{girl} \mid \text{boy}) = \frac{Prob(\text{BG or GB})}{Prob(\text{not GG})} = \frac{2/4}{3/4} = \frac{2}{3}$$

$\rightarrow$ answer c.

Example

Example 3

— In a population X, 1% of the people have a disease.

— A test has accuracy 95% : whether or not you have the disease, the test gives you the truth with probability 95%, and the opposite of the truth with probability 5%.

— Question 1 : *before taking the test*, what's the probability that you have the disease ?

— Question 2 : after a positive test result, what's the probability that you have the disease ? a. more than 90%, b. between 30% and 90%, c. less than 30%

Understand why : suppose there are 2000 people in the population and everybody gets tested. How many people really have the disease ? How many of the healthy people receive a positive test result ? How many of the sick ? What fraction of the people who receive a positive test result are actually sick ?

Solution (not on the slides) :

— Question 1 : 1%.

— 2000 people : 20 sick, 1980 healthy.

— Healthy with a positive test : $5\% \times 1980 = 99$.

— Sick with a positive test : $95\% \times 20 = 19$.

$$Prob(\text{sick} \mid +) = \frac{19}{19+99} = \frac{19}{118} \approx 16\%$$

$\rightarrow$ answer c (less than 30%).

Weekly Summary

Summary

— **Braess paradox** : adding a superfast route moves the NE from 65 to 80 minutes.

— Every network has a pure-strategy NE : the minimum of **potential energy**, found by improving one player at a time.

— Bayesian games : free riding with two types of Player 2, Harsanyi representation.

— **Continuum of types** : threshold strategies, $T_1^* = T_2^* = 2/3$.

— **Conditional probability** : $Prob(A|B) = Prob(A\&B)/Prob(B)$, used to update probabilities with new information.